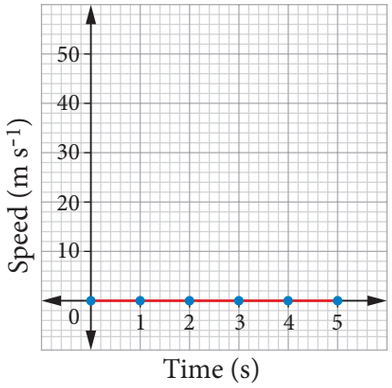
a. Bus at rest

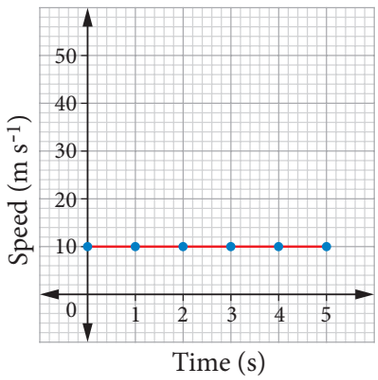
Time (s)	0	1	2	3	4	5
Speed (ms^{-1})	0	0	0	0	0	0



The speed of the bus remains at 0 ms^{-1}. So, the bus has zero acceleration.

b. Bus travelling at uniform speed of ms^{-1}

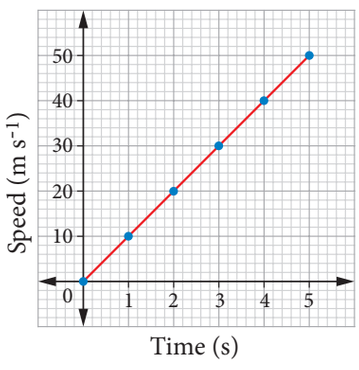
Time (s)	0	1	2	3	4	5
Speed (ms^{-1})	10	10	10	10	10	10



The speed of the bus remains at 10 ms^{-1}. Here, slope of the line is zero. So, the bus has zero acceleration.

c. Bus travelling uniform acceleration

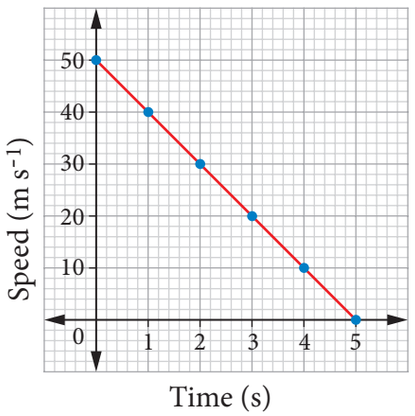
Time (s)	0	1	2	3	4	5
Speed (ms^{-1})	10	10	20	30	40	50



The speed of the bus increases by 10 ms^{-1} every second. Hence, the graph has a positive and constant gradient, and the acceleration is constant.

d. Bus travelling uniform deceleration

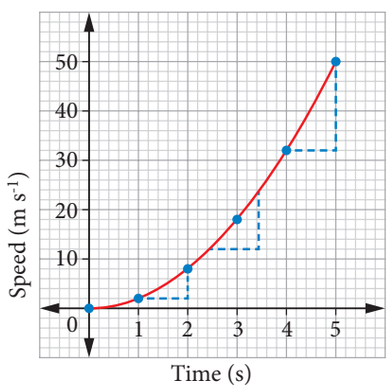
Time (s)	0	1	2	3	4	5
Speed (ms^{-1})	50	40	30	20	10	0



The speed of the bus decreases by 10 ms^{-1} very second. Hence, the graph has a negative and constant gradient and the acceleration is negative and constant.

e. Bus travelling with increasing acceleration (Non-uniform acceleration)

Time (s)	0	1	2	3	4	5
Speed (ms^{-1})	10	2	8	18	32	50



The speed of the bus is increasing with time. Hence, the graph has a positive and increasing gradient and the acceleration increases.

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